

Tollgate

Enterprise Runtime Governance for AI Agents

Internal Technical Review

Executive Summary

Tollgate is a lightweight, framework-agnostic runtime enforcement layer for AI agents. As AI agents become increasingly autonomous in enterprise systems, Tollgate provides the governance, security, and visibility needed for production deployments.

Capability	Description
Policy Enforcement	YAML-based rules with ALLOW/DENY/ASK decisions
Security Controls	Encryption, signatures, rate limiting, network guards
Policy Versioning	Git-like version control with rollback
SLO Monitoring	Availability, latency, and error rate tracking
Reputation System	Dynamic trust scores with adaptive limits
Workflow Orchestration	Multi-step approval chains and escalation
Observability	OpenTelemetry tracing, metrics, structured logging

The Challenge

- AI agents are becoming autonomous decision-makers in enterprise systems
- Unrestricted tool access creates security, compliance, and operational risks
- Traditional API gateways don't understand agent intent or context
- No standardized way to enforce policies across different agent frameworks
- Lack of visibility into what agents are doing and why

The Solution

- A lightweight, framework-agnostic runtime enforcement layer
- Policy-as-code governance with YAML-based rule definitions
- Real-time decision making: ALLOW, DENY, or escalate to human approval
- Complete audit trail with cryptographic verification
- Integrates with any agent framework: LangChain, CrewAI, AutoGen, MCP, and more

Enterprise Security

- AES-256-GCM field-level encryption for sensitive audit data
- SHA-256 hash chains for immutable, tamper-evident audit logs
- Ed25519 signatures for cryptographic agent context verification
- Configurable rate limiting to protect against abuse
- Domain allowlists and network request filtering

Policy Versioning & Rollback

- Complete version history with author and timestamp tracking
- Instant rollback to any previous policy version
- Diff comparison showing exact changes between versions
- Hot reload capability - update policies without restarts
- Duplicate detection to prevent redundant versions

SLO Monitoring & Alerting

- Availability SLOs for system uptime tracking
- Latency percentile tracking (P50, P95, P99)
- Error rate monitoring with configurable thresholds
- Approval/denial rate tracking for governance metrics
- Alert deduplication and automatic recovery detection

Agent Reputation System

- Dynamic 0.0-1.0 trust scores based on behavior
- Tracks successes, failures, policy violations, anomalies
- Automatic rate limit adjustment based on trust level
- Time-based score decay toward configurable baseline
- Integration with audit sink for automatic updates

Workflow Orchestration

- Multi-step sequential or parallel approval chains
- Conditional routing based on context (risk level, amount, etc.)
- Automatic escalation paths with configurable timeouts
- Pre-built workflow templates for common patterns
- SQLite and in-memory storage backends

Demo Scenarios

The following scenarios can be demonstrated to showcase Tollgate's capabilities:

1. Policy Enforcement Demo

Show how different tool calls receive ALLOW, DENY, or ASK decisions based on policy rules. Demonstrate pattern matching, context conditions, and grant-based overrides.

2. Human Approval Workflow

Walk through a sensitive operation that requires human approval. Show the approval queue, the decision UI, and how the agent receives and processes the approval response.

3. Reputation System in Action

Execute a series of operations that affect an agent's trust score. Show how low-trust agents receive stricter rate limits and may be blocked from sensitive operations.

4. Policy Hot Reload

Demonstrate updating a policy file and showing immediate effect on decisions without restarting the agent. Show version history and rollback capability.

5. Workflow Orchestration

Execute a multi-level approval workflow with conditional routing. Show how high-value operations route to different approval chains than standard operations.

Technical Details

Metric	Value
Test Coverage	535+ automated tests
Language	Python 3.10+
Async Support	Full async/await architecture
Storage Backends	SQLite, Redis, In-Memory
Observability	OpenTelemetry, Prometheus, JSON logs
License	MIT

Integration Example

Integrating Tollgate into an existing agent requires minimal code changes:

```
from tollgate import ControlTower, AgentContext, Intent, ToolRequest

# Initialize with policy file
tower = ControlTower(policy='policy.yaml')

# Create context and intent
context = AgentContext(agent_id='agent-1', session_id='sess-1')
intent = Intent(action='fetch_data', reason='User requested report')
request = ToolRequest(tool_name='database_query', arguments={'query': '...'})
```



```
# Execute with governance
decision = await tower.execute(context, intent, request)

if decision.effect == Effect.ALLOW:
    # Proceed with tool execution
    result = await execute_tool(request)
elif decision.effect == Effect.DENY:
    # Handle denial
    return f"Operation denied: {decision.reason}"
elif decision.effect == Effect.ASK:
    # Escalate to human approval
    approval = await request_approval(decision)
```

Recommended Next Steps

Explore the Codebase

Clone the repository and run the test suite to verify all features work correctly.

Try the CLI Tools

Use 'tollgate policy validate', 'tollgate grant list', and 'tollgate health' commands.

Integrate with Sample Agent

Create a simple agent integration to see Tollgate in action.

Discuss Production Requirements

Identify specific security, compliance, and scalability needs.

Identify Pilot Use Cases

Select 1-2 internal AI agent projects for initial adoption.

For questions or to schedule a deeper technical review, please reach out.